

Seolmin YANG

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ACADEMIC POSITION

School of Data Science University of Virginia Postdoctoral Research Associate (supervisor: Prof. Yong-Yeol Ahn)	2025.08 - present
Observatory on Social Media Luddy School of Informatics, Computing, and Engineering Indiana University Bloomington Postdoctoral Fellow (supervisor: Prof. Yong-Yeol Ahn)	2025.02 - 2025.08
Kellogg School of Management Northwestern University Postdoctoral Fellow (supervisor: Prof. Hyejin Youn)	2024.02 - 2025.02

EDUCATION

Ph.D. Science and Technology Policy KAIST, Daejeon, South Korea (Dissertation) <i>Research Innovation and Collaboration in Government-Funded Science</i> - Advisor: So Young Kim - Ph.D. Dissertation Fellowship, National Research Foundation of Korea	2024
Visiting Predoctoral Fellow , Northwestern Institute on Complex Systems, Northwestern University Supervisor: Prof. Hyejin Youn	2019.12 - 2020.03
M.S. Science and Technology Policy KAIST, Daejeon, South Korea (Thesis) <i>Understanding the Process of Knowledge Creation in University Laboratories: An Analysis of Laboratory Dynamics at KAIST</i> - Advisor: So Young Kim	2016
B.S. Chemistry KAIST, Daejeon, South Korea Minor in Chemical & Biomolecular Engineering	2014

AREAS OF INTERESTS

Technology Evolution, Science of Science, Innovation Policy, R&D Management

PEER-REVIEWED PUBLICATIONS

Yang, S. & Kim, S. Y. (2023). Knowledge-integrated research is more disruptive when supported by homogeneous funding sources: A case of US federally funded research in biomedical and life sciences. *Scientometrics*, 128, 3257-3282.

Yang, S. & Kim, S. Y. (2021). Managing individual knowledge creation with demographic faultlines: A case of university laboratories. *Asian Journal of Technology Innovation*, 29(3), 349-368.

Lee, S., **Yang, S.**, & Nah, K. (2020). Discovering the topology of design process: A network analysis of author keywords in Design Studies. *Archives of Design Research*, 33(1), 191-203.

WORKS IN PROGRESS

Yang, S., & Youn, H. (2023). Geometrics of the adjacent possible: Harvesting values at the curvature. *arXiv preprint arXiv:2311.16360*.

Yang, S., Jeong, D., & Youn, H. Evolution of technology landscape from micro- to macro-structures.

Yang, S., Youn, H., & Jung, H. Boundary Reclassification.

Jeong, D., **Yang, S.**, Evans, J., & Youn, H. Exploring consolidation pathways in knowledge assemblies.

Jeong, D., **Yang, S.**, Jang, S., & Shin, S. R. Links from papers to patents are becoming less disruptive than the reverse.

AWARDS & FELLOWSHIPS

Best Paper Award, Summer 2022 Korea Society Innovation Management & Economics Academic Conference	2022.07
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Title: Reducing the risk of exploration: Innovation relying on government-funded research

Ph.D. Dissertation Fellowship, National Research Foundation of Korea	2021.06 - 2022.05
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Title: Making 1+1 more than 2: A scientometric analysis of the efficacy of government funding for interdisciplinary research

Emerging Scholar Award, 17th International Conference on Knowledge, Culture, and Change in Organizations, Darwin, Australia	2017
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STEPI Fellowship, Science and Technology Policy Institute, South Korea	2017
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Title: Understanding the organization of team science: A case of the institute for basic science in South Korea

PEER-REVIEWED CONFERENCE PRESENTATIONS

- Yang, S.,** Jeon, D., Ahn, Y.-Y., & Youn, H. (2026, June). *Idea Timing: Structure as the Hidden Variable in Innovation*. Contributed Talk, NetSci 2026, Boston, MA.
- Yang, S.** (2025, November). *Mapping Innovation: Geometrics of the Adjacent Possible*. Poster, Datapalooza 2025, School of Data Science, University of Virginia, Charlottesville, VA.
- Yang, S.** (2023, June). *Skewed Funding for Basic Sciences: Unpacking the Basicness of NIH Grants*. Poster, International Conference on the Science of Science and Innovation (ICSSI), Evanston, IL.
- Kwon, H. Y., **Yang, S.,** & Youn, H. (2022, July). *Occupational Mobility and Wages from a Perspective of Worker Skills*. Poster, IC2S2, Chicago, IL.
- Yang, S.,** Youn, H., & Kim, S. Y. (2022, July). *Reducing the Risk of Exploration: Innovation Relying on Government-Funded Research*. Paper, Summer 2022 Korea Society Innovation Management & Economics Academic Conference, Jeju, South Korea.
- Yang, S.,** & Kim, S. Y. (2019, October). *Strategizing Scientific Awards: The Impacts of Early-Career Awards on the Research Network of Recipient Scientists*. Poster, 8th Biennial Atlanta Conference on Science and Innovation Policy, Atlanta, GA.
- Yang, S.,** & Kim, S. Y. (2017, October). *Re-combining Social Category Diversity and Tenure Diversity: A Cross-Level Analysis of Individual Explicit Knowledge*. Paper, 7th Biennial Atlanta Conference on Science and Innovation Policy, Atlanta, GA.
- Yang, S.** (2017, September). *Affiliation Diversity and Team Performance: Impact of the External Social Network Ties of Leading Scientists*. Paper, 15th International Triple Helix Conference, Daegu, South Korea.
- Yang, S.,** & Kim, S. Y. (2017, April). *Re-combining Social Category Diversity and Tenure Diversity: A Cross-Level Analysis of Individual Explicit Knowledge Creation*. Paper, 17th International Conference on Knowledge, Culture, and Change in Organizations, Darwin, Australia.
- Yang, S.,** & Kim, S. Y. (2016, September). *Understanding the Process of Knowledge Creation in University Laboratories: An Analysis of Laboratory Dynamics at KAIST*. Paper, XIV International Triple Helix Conference, Heidelberg, Germany.

INVITED PRESENTATIONS & TALKS

- Yang, S.** (2017, November). *Demographic faultlines and psychological safety: A cross-level analysis of individuals' reliance on explicit and tacit knowledge*. Talk given at the Seminar on Innovation for Global Development (led by Prof. Calestous Juma), Harvard Kennedy School, Cambridge, MA.
- Yang, S.** (2017, June). *Re-combining social category diversity and tenure diversity: A cross-level analysis of individual explicit knowledge creation*. Talk given at the Joint S&T Policy Workshop, School of Public Policy & Management, Tsinghua University, Beijing, China.

EXPERIENCE**Teaching Assistant**

Spring 2020

Winning with Networks (MORS-457-0), Northwestern University (Instructor: Prof. Hye-jin Youn)

Role: assisted students in handling and visualizing network data.

Research Assistant

2019

The Trend of Artificial Intelligence in South Korea, KAIST-Fourth Industrial Revolution Center (FIRIC), Daejeon, South Korea (PI: Prof. So Young Kim)

Role: analyzed bibliometric data about AI and conducted network analysis.

Research Assistant

2018

A Proposal for Improvement of the Current Science Awards System with Its Comprehensive Assessment in South Korea, Korea National Research Foundation, Daejeon, South Korea (PI: Prof. So Young Kim)

Role: analyzed survey data to assess factors affecting scientists' perceptions of science awards.

Head Research Assistant

2018

A Study of Future Technology Roadmaps based on the Time Series Analysis of Technologies, Korea National Research Foundation, Daejeon, South Korea (PI: Prof. So Young Kim)

Role: conducted and interpreted interviews about organizations' processes for reviewing, selecting, and publishing emerging technologies; analyzed trends of emerging technologies from public institutions in South Korea over a decade.

Head Research Assistant

2017

A Policy for Collaborative Lab Design, KAIST (PI: Prof. Hongjin Kong)

Role: analyzed university laboratory systems in Japan, Germany, and France; interpreted interviews with lab heads about research collaboration.

Head Research Assistant

2016

Evidence-based Appraisal of the Program for Women Returners to Research & Development, Center for Women In Science, Engineering and Technology, Seoul, South Korea (PI: Prof. So Young Kim)

Role: conducted and analyzed focus-group interviews with female scientists and engineers returning to the profession; analyzed survey data to assess factors affecting successful return.

Research Assistant

2016

The Impact of IBS on Regional Development, Institute for Basic Science, Daejeon, South Korea (PI: Prof. Yoonjung Lee)

Role: analyzed survey data to measure the market values of non-market goods or services.

Researcher, Center for Strategic Foresight, Science and Technology Policy Institute, Seoul, South Korea.

2013 - 2014

STATISTICAL SKILLS

R, Python, Gephi, UCINET, SPSS